Date: November 7, 2026

UNIVERSITY OF MOHAGHEGH ARDABILI

PROJECT PROPOSAL

Smart Real Estate & Consultation System (Powered by AI Agent)

PROJECT INFORMATION	DEVELOPMENT TEAM
Course: Software Lab & Internet Engineering **Instructor:** Dr. Mehdi Molaei **Team Name:** Mohaghegh	• **Amirreza Soleimani** • **Iliya Mehri**

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1. Team Name & Members

University: University of Mohaghegh Ardabili
Instructor: Dr. Mehdi Molaei

Team Member Roles

Member Name	Role
Amirreza Soleimani	Front-End
Iliya Mehri	Back-End

2. Project Vision

This system is a comprehensive, smart web platform designed to facilitate and modernize property buying, selling, renting, and leasing processes. The primary goal of this platform is to elevate the traditional, manual property search experience by incorporating an interactive **AI Agent**. By analyzing user requirements, personal preferences, budget constraints, and lifestyle criteria, the AI Agent recommends tailored property options and acts as a 24/7 dedicated consultant throughout the decision-making journey.

3. Target Audience

- **Buyers & Tenants:** Individuals seeking property matching their budget and lifestyle requirements who wish to leverage intelligent recommendations.
- **Sellers & Property Owners:** Property owners looking to list, price, manage, and market their properties efficiently.

- **Real Estate Agents & Agencies:** Industry professionals requiring a unified management dashboard to handle property portfolios and connect with potential clients.
- **System Administrators:** Administrators responsible for overseeing platform activities, verifying user credentials, and approving listing submissions.

4. Core Features / User Stories

4.1. Buyers & Tenants

1. **Registration & Profile Management:** As a property seeker, I want to create an account so that I can save my search filters and favorite listings.
2. **Advanced Search & Filtering:** As a buyer/tenant, I want to filter listings by price, area, bedroom count, location, and deed type so that I can quickly pinpoint suitable options.
3. **Property Detail View:** As a property seeker, I want to view detailed listing specs, high-resolution photo galleries, map locations, and contact info so that I can make informed decisions.
4. **Smart Comparison:** As a buyer, I want to compare selected properties side-by-side in a comparative matrix so that I can evaluate key differences easily.
5. **Direct Communication:** As a tenant/buyer, I want to message listing agents directly via in-app chat so that I can inquire about properties or schedule viewings.
6. **Interactive AI Consultation:** As a buyer, I want to converse with the AI Agent about my budget and preferences (e.g., proximity to

metro or schools) so that I receive personalized property recommendations.

7. **AI Price Valuation:** As a buyer, I want the AI Agent to evaluate whether a property's asking price aligns with regional market averages so that I can ensure fair pricing.

4.2. Sellers & Property Owners

1. **Listing Creation & Publishing:** As a property owner, I want to upload property specs, photos, and prices so that my listing is visible to prospective buyers and tenants.

2. **Listing Management:** As a seller, I want to edit details, update pricing, or change status (e.g., Active/Sold) so that my property information remains accurate.

3. **AI Pricing Guidance:** As a property owner, I want pricing suggestions from the AI Agent so that I can price my property competitively and increase sell-through speed.

4.3. Real Estate Agents & Agencies

1. **Portfolio Dashboard:** As a real estate agent, I want a dedicated dashboard to manage multiple listings simultaneously so that I can streamline my property inventory.

2. **AI Lead Generation:** As an agent, I want the system to match buyer inquiries with my property listings so that I can connect directly with high-intent leads.

3. **Featured Listings:** As an agent, I want to promote specific listings to the top of search results and AI recommendation feeds to maximize visibility.

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4.4. System Administrators

1. **Listing Review & Moderation:** As an admin, I want to review and approve/reject new property listings prior to publication to prevent fraudulent content.
2. **User Management & Verification:** As an admin, I want to manage user permissions and suspend accounts in case of policy violations.
3. **AI Analytics & System Oversight:** As an admin, I want to monitor AI Agent usage statistics and user feedback metrics to refine recommendation algorithms.

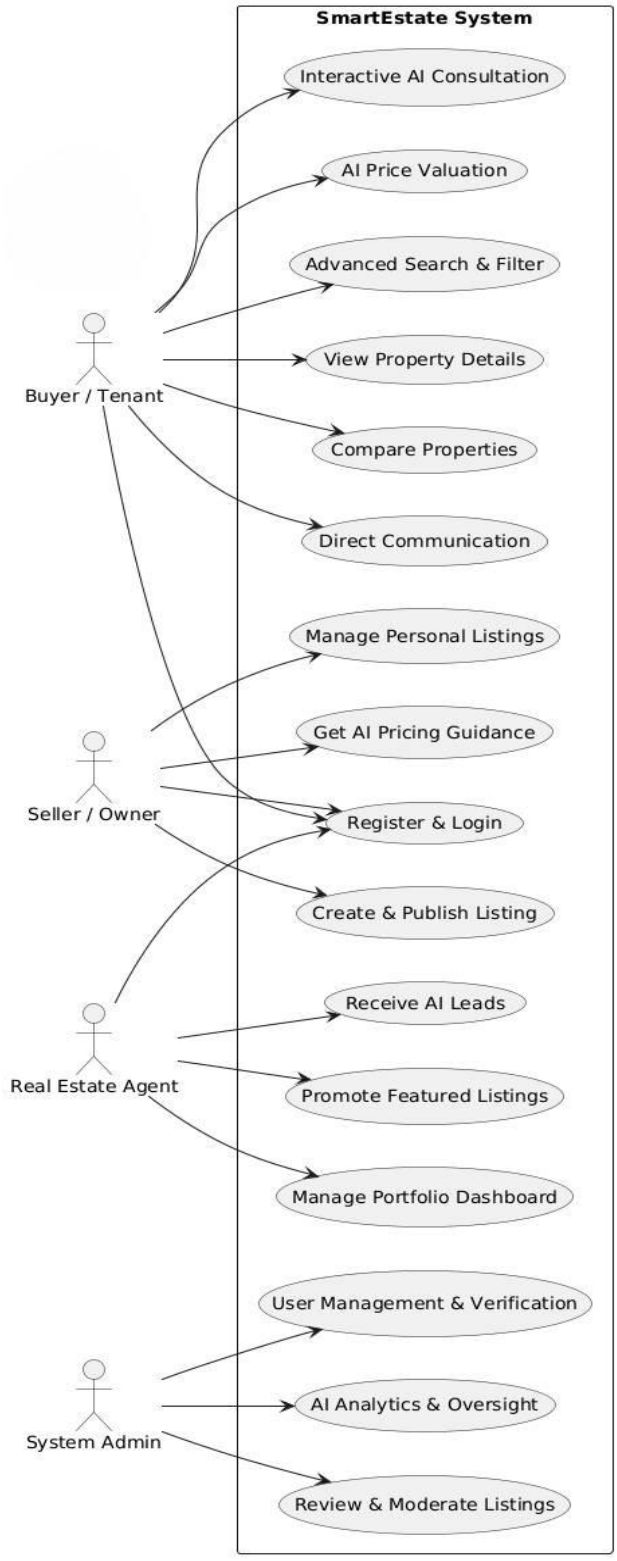
5. Links

- **GitHub Repository:** https://github.com/Iliya-Mehri/Software-Engineering-Lab-Project/tree/main
- **Project Management Board (Mizito):** https://office.mizito.ir/#/ws/projects/dialog/6ad322ff848f35231072f45b/

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6. System Modeling & UML Diagrams

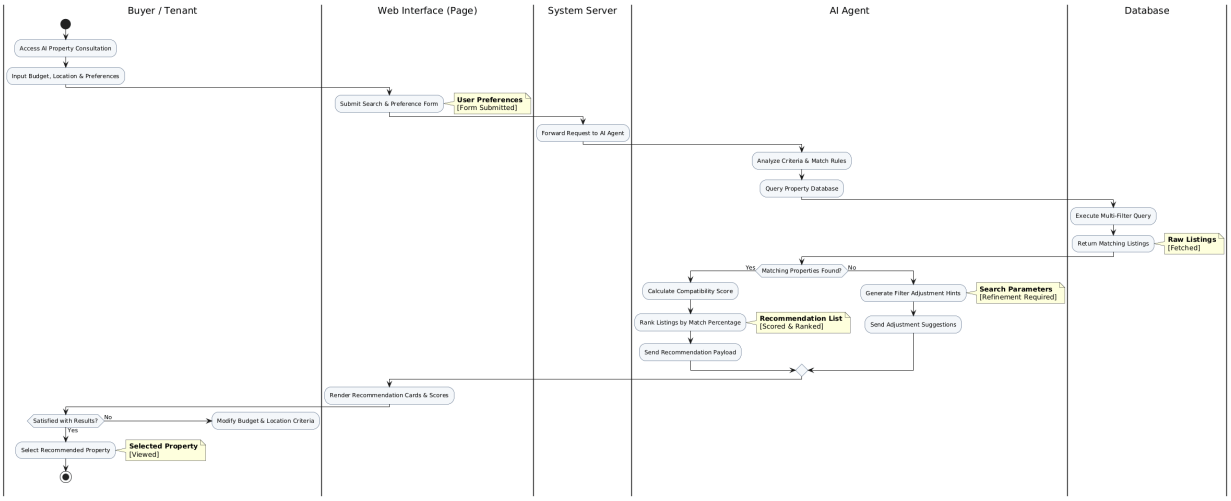
6.1. Use Case Diagram



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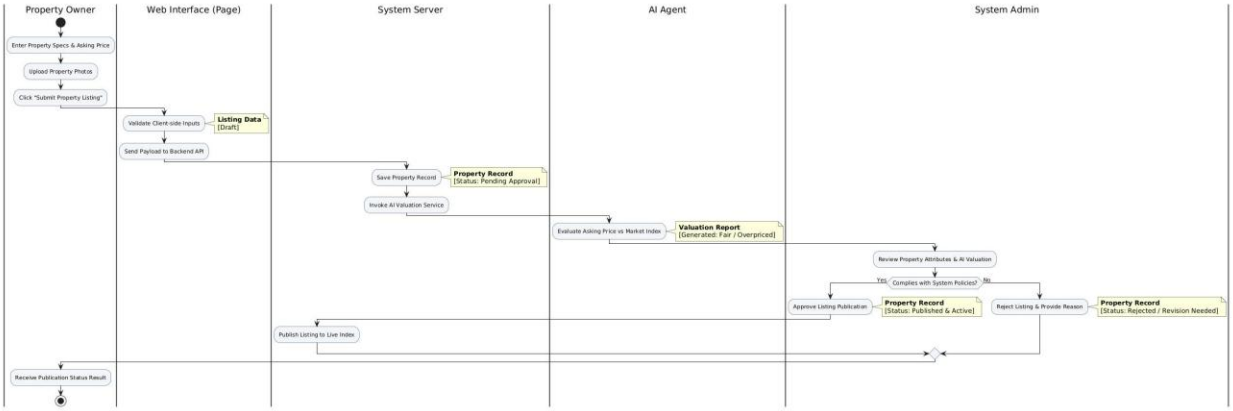
6.2. Activity Diagrams

6.2.1. Activity Diagram 1: AI Recommendation



This diagram details the sequence of actions when a buyer requests AI-driven property recommendations, from initial preference submission to database querying and score ranking.

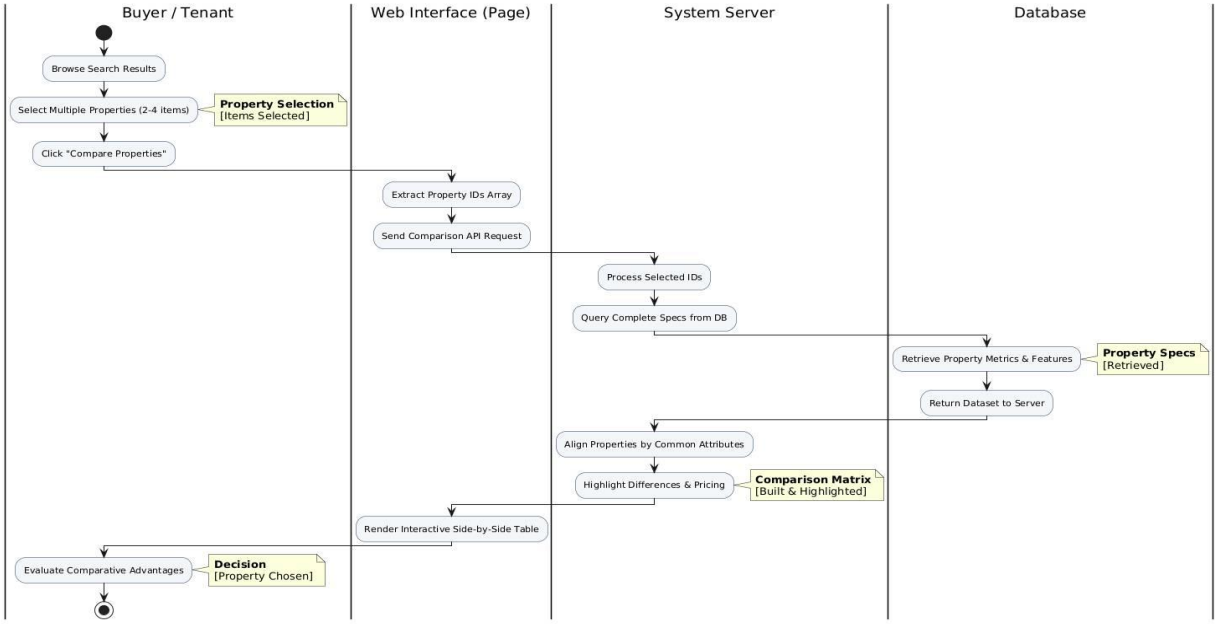
6.2.2. Activity Diagram 2: Create & Moderate Property Listing



This diagram outlines the end-to-end lifecycle of listing creation by a property owner, automated valuation by the AI engine, and final manual approval by the System Administrator.

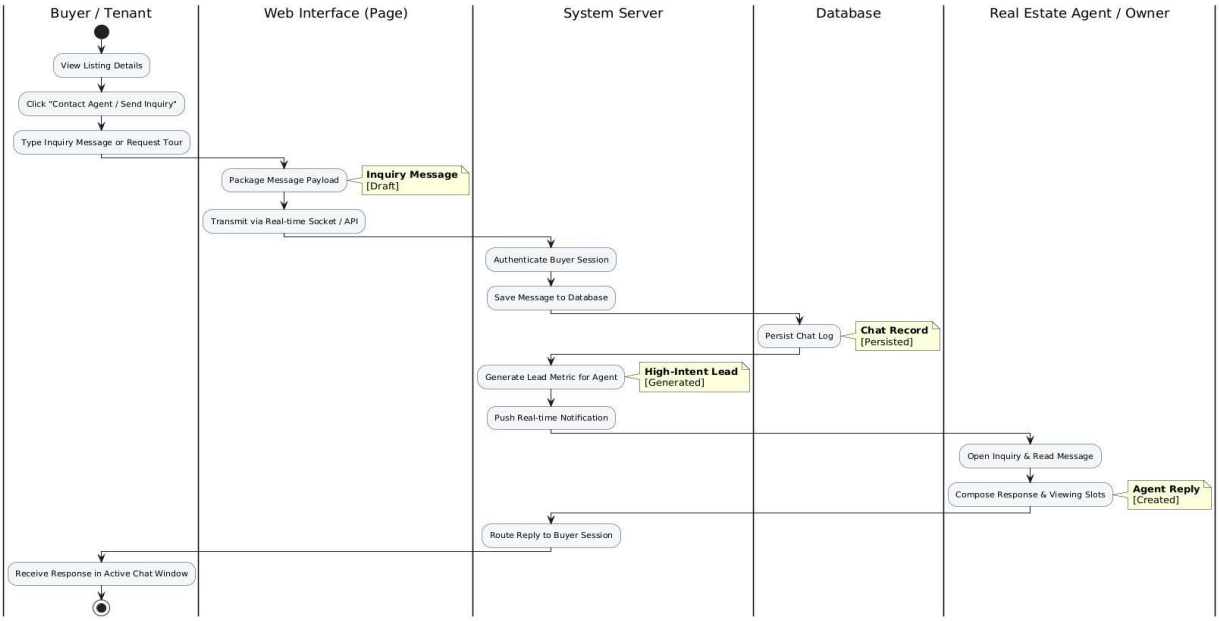
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6.2.3. Activity Diagram 3: Smart Property Comparison



This workflow illustrates how the web interface fetches multi-property parameters from the database to generate a comparative analysis matrix for prospective buyers.

6.2.4. Activity Diagram 4: Direct Communication & Messaging

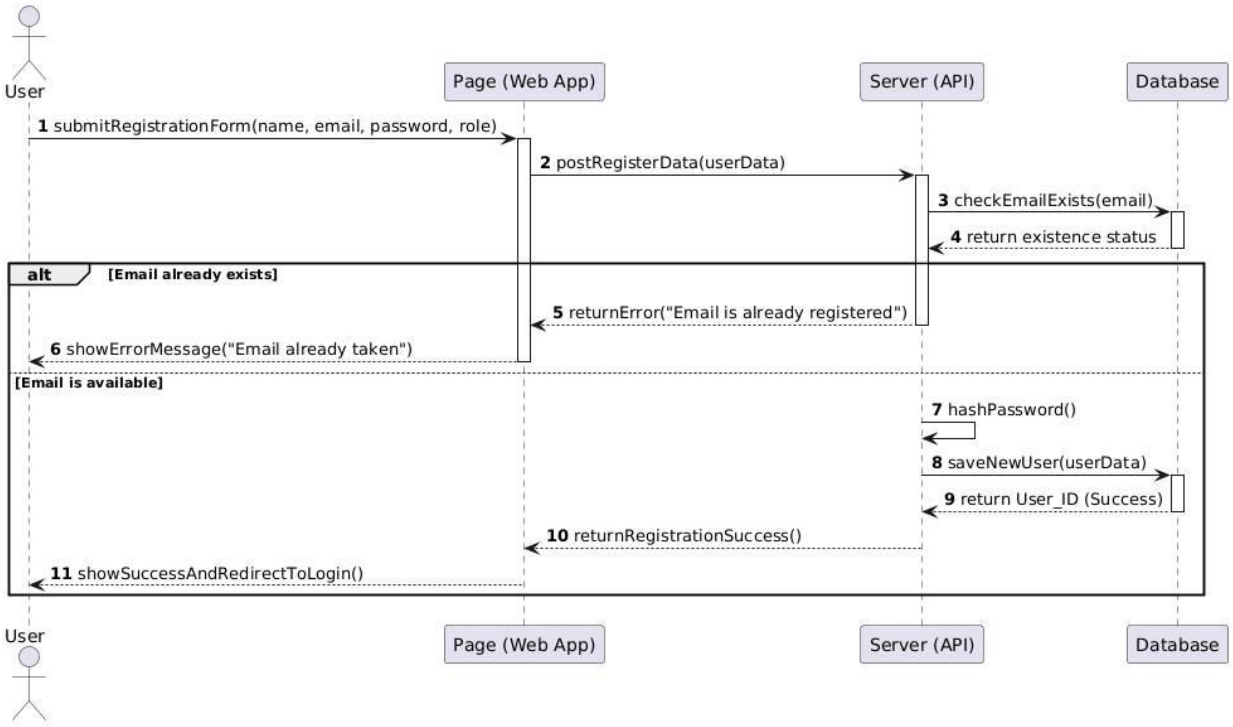


This diagram models the real-time messaging pipeline between buyers and real estate agents/owners, including message persistence and push notifications.

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6.3. Sequence Diagrams

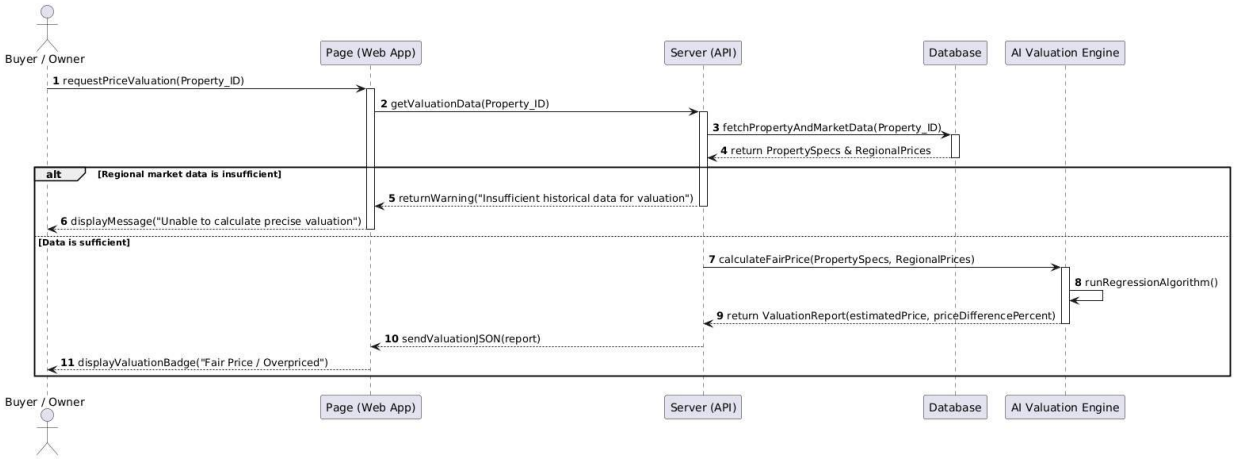
6.3.1. Sequence Diagram 1: User Registration & Authentication



This diagram models the secure registration workflow. It handles email availability verification using conditional `alt` blocks, securely hashes user passwords, and persists new account credentials in the database.

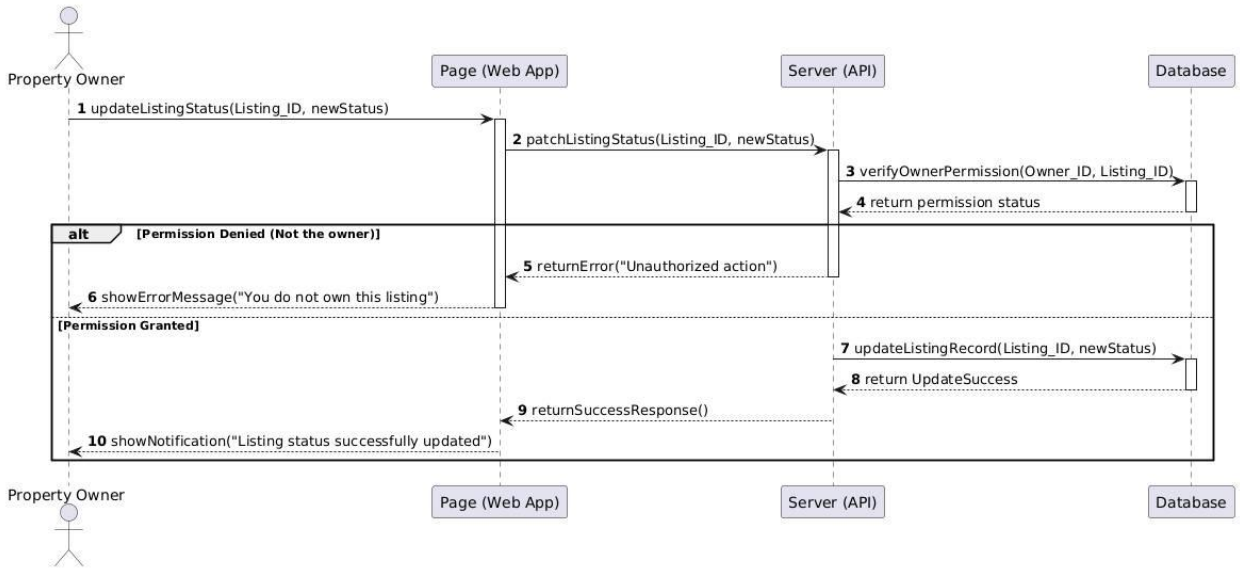
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6.3.2. Sequence Diagram 2: AI Price Valuation



Demonstrates the interaction pipeline when a user requests an automated price estimation for a specific listing. The system checks historical regional data availability via conditional logic before executing the AI regression algorithm.

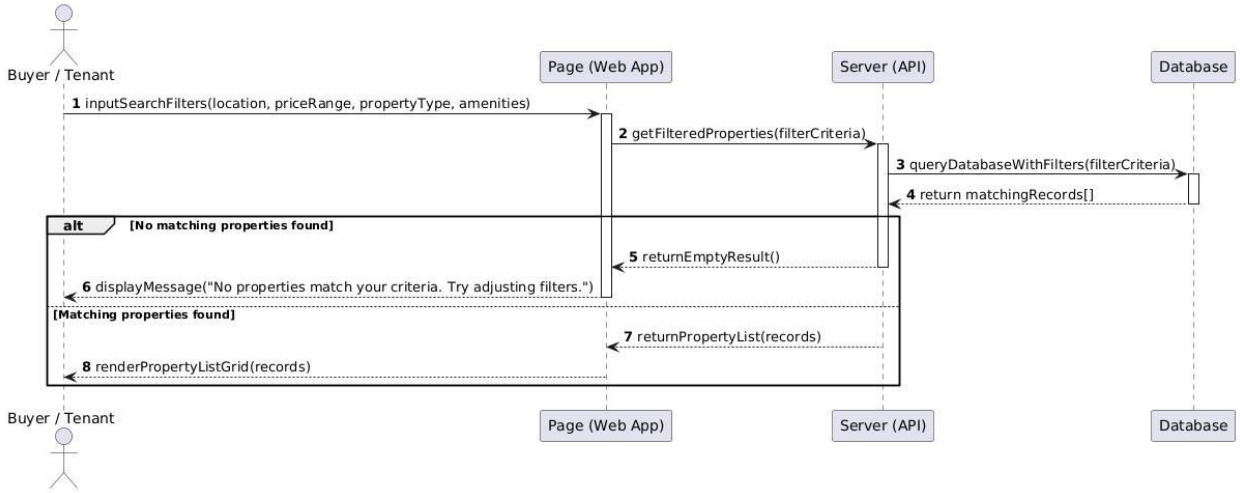
6.3.3. Sequence Diagram 3: Listing Management & Status Update



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Shows how a property owner updates a listing's status (e.g., setting it to "Sold"). The server validates ownership permissions prior to applying updates to the database, handling unauthorized attempts through an alternative flow.

6.3.4. Sequence Diagram 4: Advanced Property Search & Filtering



Models the dynamic search workflow where users submit multi-criteria filters. The diagram handles edge cases using conditional logic to gracefully notify users when no matching listings are found in the database.